Understanding learning rates as weighted sums

Session 1: deriving Equation 6, step by step

Where we start

In session 1 the model learns the probability that green will be rewarded with two equations. On each trial it calculates a prediction error,

$$\underbrace{\delta_t}_{\text{prediction error}} = \underbrace{o_t}_{\text{outcome}} - \underbrace{p_t}_{\text{model prediction}} \qquad \text{(Equation 1)}$$

and uses it to update its expectation:

$$\underbrace{p_{t+1}}_{\text{new prediction}} = \underbrace{p_t}_{\text{old prediction}} + \underbrace{\alpha\delta_t}_{\text{scaled prediction error}} \qquad \text{(Equation 2)}$$

Here o_t is the outcome on trial t (1 if green was rewarded, 0 if orange was rewarded), p_t is the current estimate of the probability that green will be rewarded, and α is the learning rate, with $0 < \alpha \leq 1$.

These equations only ever look one trial back. Yet the notebook claims that the prediction depends on *every* past outcome, as a weighted sum:

$$p_{T+1} = (1-\alpha)^T p_1 + \sum_{t=1}^{T} w_t o_t \qquad \text{(Equation 5)}$$

with weights

$$w_t = \alpha(1-\alpha)^{T-t} \qquad \text{(Equation 6)}$$

where T is the current trial and $T - t$ is how many trials into the past we are looking. This handout shows where these two equations come from.

Step 1: Combine Equations 1 and 2 into one

Put the prediction error from Equation 1 into Equation 2, and collect the two terms that contain p_t:

$$\begin{aligned} p_{t+1} &= p_t + \alpha(o_t - p_t) \\ &= p_t - \alpha p_t + \alpha o_t \\ &= (1-\alpha)\, p_t + \alpha o_t. \end{aligned}$$

This already says something useful: the new prediction is a *mix* of the old prediction and the newest outcome. With $\alpha = 0.3$, for example, it is 70% old prediction and 30% new outcome.

Step 2: Apply the rule trial by trial

The model starts with a prediction p_1 on the first trial (`startingProb`, 0.5 by default). Now apply the rule from Step 1 over and over. Each time, replace the previous prediction with what the line above already worked out for it.

After one trial:

$$p_2 = (1-\alpha)\, p_1 + \alpha o_1.$$

After two trials:

$$\begin{aligned} p_3 &= (1-\alpha)\, p_2 + \alpha o_2 \\ &= (1-\alpha)\, \underbrace{\left[(1-\alpha)\, p_1 + \alpha o_1\right]}_{p_2} + \alpha o_2 \\ &= (1-\alpha)^2 p_1 + \alpha(1-\alpha)\, o_1 + \alpha o_2. \end{aligned}$$

After three trials:

$$\begin{aligned} p_4 &= (1-\alpha)\, p_3 + \alpha o_3 \\ &= (1-\alpha)\, \underbrace{\left[(1-\alpha)^2 p_1 + \alpha(1-\alpha)\, o_1 + \alpha o_2\right]}_{p_3} + \alpha o_3 \\ &= (1-\alpha)^3 p_1 + \alpha(1-\alpha)^2 o_1 + \alpha(1-\alpha)\, o_2 + \alpha o_3. \end{aligned}$$

A pattern is appearing. Read the last line from right to left:

- the outcome of the *last* trial, o_3, is multiplied by α;
- the outcome one trial before that, o_2, by $\alpha(1-\alpha)$;
- the outcome two trials before that, o_1, by $\alpha(1-\alpha)^2$;
- and the starting prediction p_1 by $(1-\alpha)^3$.

Each step back in time adds one more factor of $(1-\alpha)$.

Step 3: The pattern holds on every trial

The pattern carries on for any number of trials, and the reason is simple. Every new trial does the same two things: it multiplies *everything already in the prediction* by $(1-\alpha)$, and it adds the newest outcome multiplied by α. So an outcome from one trial further back has picked up exactly one more factor of $(1-\alpha)$.

After T trials, the outcome of trial t lies $T-t$ trials in the past, so it has picked up $T-t$ factors of $(1-\alpha)$, and the starting prediction has picked up T of them. That is Equations 5 and 6:

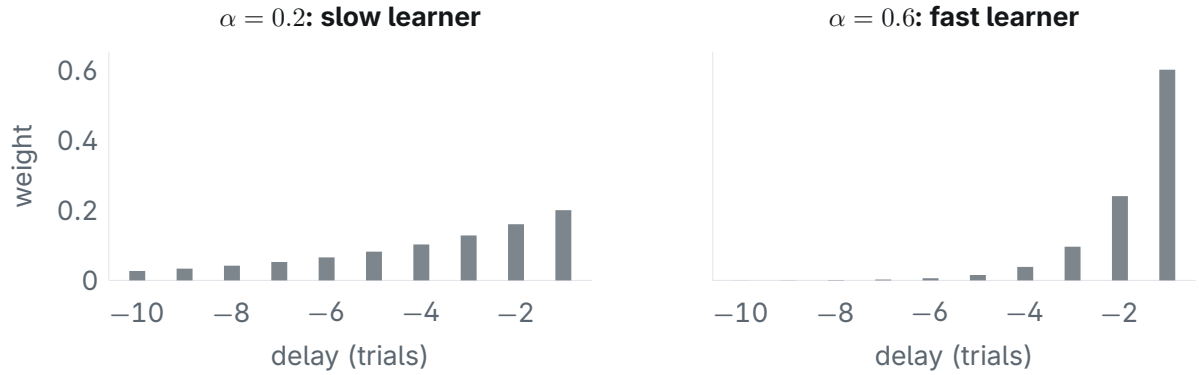
$$p_{T+1} = \underbrace{(1-\alpha)^T p_1}_{\substack{\text{what is left of the} \\ \text{starting prediction}}} + \sum_{t=1}^{T} \underbrace{\alpha(1-\alpha)^{T-t}}_{w_t}\, o_t$$

The symbol $\sum$ means "add up". Here it says: for every trial t from 1 to T, multiply its outcome by its weight w_t, and add the results together. Written out for $T = 3$, it is exactly what we found for p_4:

$$\sum_{t=1}^{3} \alpha(1-\alpha)^{3-t} o_t = \alpha(1-\alpha)^2 o_1 + \alpha(1-\alpha)\, o_2 + \alpha o_3.$$

What this tells us

Older outcomes count less. Because $1 - \alpha$ is less than 1, each step back multiplies the weight by the same factor $1 - \alpha$, so the weights shrink steadily. The last trial carries weight α.



A fast learner (large α) bases its prediction almost entirely on the last few outcomes. It adapts quickly when the reward probability changes, but every chance outcome throws it around. A slow learner (small α) spreads its weight over a long stretch of the past. Its prediction is steadier, but it takes longer to catch up after a change. With $\alpha = 1$, only the last outcome counts.

The starting prediction is forgotten. p_1 is multiplied by $(1 - \alpha)^T$, which shrinks towards zero as trials go by. With $\alpha = 0.2$, after 20 trials only $0.8^{20} \approx 0.01$ of it is left, so the starting probability hardly matters once there are enough trials.

The weights add up to 1. Suppose the starting prediction is $p_1 = 1$ and green is rewarded on every trial. Then every prediction error is $1 - 1 = 0$, so the prediction never moves and stays at 1. In Equation 5, every o_t and p_1 is then 1, so the right-hand side is just the sum of all the weights. Since the left-hand side is 1,

$$(1 - \alpha)^T + \sum_{t=1}^{T} w_t = 1.$$

The prediction is therefore a weighted *average* of zeros and ones, which is why it always stays between 0 and 1, as a probability must.